Step1: proc.h

```
  int sched_count;
  int run_ticks;
```

Step2:proc.c

```
    p->sched_count++;
```

Step3:trap.c

```
   if(myproc() && myproc()->state==RUNNING)
    {
      myproc()->run_ticks++;
    }
```

Step4: syscall.h

```
   #define SYS_getstats 28
```

Step5: syscall.c

```
   extern int sys_getstats(void);

  [SYS_getstats] sys_getstats,
```

Step6: sysproc.c

```
 int
sys_getstats(void)
{
   int *user_stats_ptr;

   if(argptr(0,(void*)&user_stats_ptr, 2*sizeof(int))<0)
   return -1;
   struct proc *p = myproc();
   int kernel_stats[2];
   kernel_stats[0]=p->sched_count;
   kernel_stats[1]=p->run_ticks;

   if(copyout(p->pgdir, (uint)user_stats_ptr,(char*)kernel_stats,
         sizeof(kernel_stats))<0)
   return -1;
   return 0;
}
```

Step 7: user.h

```
struct procstats{
  int sched_count;
  int run_ticks;
};
```

```c
int getstats(int *stats_array);
```

Step 8: usys.S

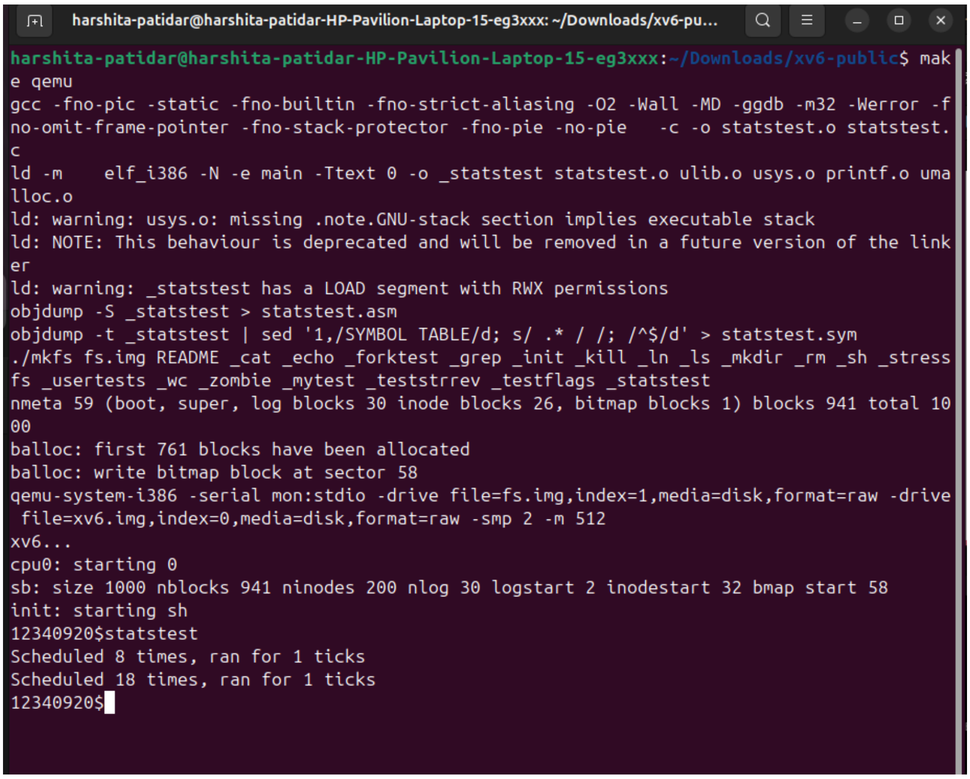
```
 SYSCALL(getstats)
```

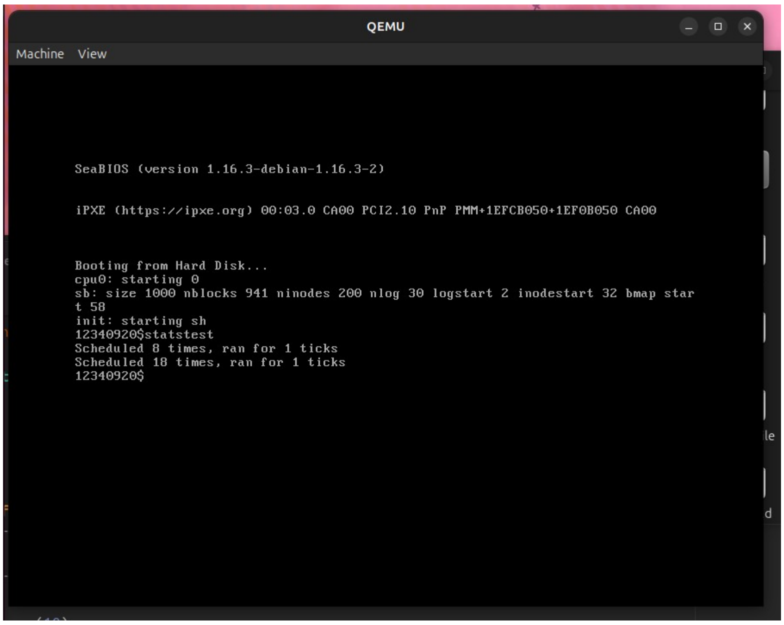
Step 8: statstest.c

```c
#include "types.h"
#include "stat.h"
#include "user.h"

int main ( void )
{
   int stats [2];
   int i ;
    for ( i = 0; i < 2; i ++) {

 if (getstats(stats) == 0) {
 printf (1 ,"Scheduled %d times, ran for %d ticks \n",stats [0],stats[1]) ;
 } else {
 printf (2 , "getstats failed \n") ;
 }
sleep (10) ;
}
exit ();
}
```

Step 9: Makefile

```
 _statstest
```


harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx: ~/Downloads/xv6-pu...
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx:~/Downloads/xv6-public$ make qemu
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -Werror -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -c -o statstest.o statstest.c
ld -m elf_i386 -N -e main -Ttext 0 -o _statstest statstest.o ulib.o usys.o printf.o umalloc.o
ld: warning: usys.o: missing .note.GNU-stack section implies executable stack
ld: NOTE: This behaviour is deprecated and will be removed in a future version of the linker
ld: warning: _statstest has a LOAD segment with RWX permissions
objdump -S _statstest > statstest.asm
objdump -t _statstest | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > statstest.sym
./mkfs fs.img README _cat _echo _forktest _grep _init _kill _ln _ls _mkdir _rm _sh _stressfs _usertests _wc _zombie _mytest _teststrrev _testflags _statstest
nmeta 59 (boot, super, log blocks 30 inode blocks 26, bitmap blocks 1) blocks 941 total 1000
balloc: first 761 blocks have been allocated
balloc: write bitmap block at sector 58
qemu-system-i386 -serial mon:stdio -drive file=fs.img,index=1,media=disk,format=raw -drive file=xv6.img,index=0,media=disk,format=raw -smp 2 -m 512
xv6...
cpu0: starting 0
sb: size 1000 nblocks 941 ninodes 200 nlog 30 logstart 2 inodestart 32 bmap start 58
init: starting sh
12340920$statstest
Scheduled 8 times, ran for 1 ticks
Scheduled 18 times, ran for 1 ticks
12340920$


QEMU
Machine View
SeaBIOS (version 1.16.3-debian-1.16.3-2)
iPXE (https://ipxe.org) 00:03.0 CA00 PCI2.10 PnP PMM+1EFCB050+1EF0B050 CA00
Booting from Hard Disk...
cpu0: starting 0
sb: size 1000 nblocks 941 ninodes 200 nlog 30 logstart 2 inodestart 32 bmap start 58
init: starting sh
12340920$statstest
Scheduled 8 times, ran for 1 ticks
Scheduled 18 times, ran for 1 ticks
12340920$